

Does a fidelity-kernel quantum SVM outperform classical baselines on malware family attribution?

A controlled comparison on two corpora

2 August 2026

Abstract

Background. Quantum kernel methods are often reported as competitive with classical machine learning on security data, but the two arms of such a comparison frequently differ in the rows they are scored on, the input dimensionality they receive, or the metric used to report the outcome.

Methods. We compare a fidelity-kernel quantum support vector machine (QSVM) against four tuned classical models on CIC-MalMem-2022 and EMBER 2018, each with a binary and a 15-class task. Both frameworks consume one shared feature pipeline whose final PCA dimensionality simultaneously sets the classical feature count and the quantum qubit budget. The quantum runner writes its stratified 1000-row subsample and its five outer folds to disk and the classical runner replays them, so both are scored on identical rows. Differences are assessed with a paired t -test over per-fold macro-F1.

Results. On the binary task of CIC-MalMem every method exceeds 0.9998 macro-F1 on the full data, so that task cannot rank methods. On the three configurations that do discriminate, classical models lead by 0.085 to 0.274 macro-F1, and all 84 comparable pairs reach $p < 0.05$. The QSVM falls to the chance level on CIC 15-class. An RBF control built on the identical feature matrix produces the same off-diagonal Gram spread on both corpora (0.2212 against 0.2184), whereas the fidelity kernel matches RBF on EMBER ($0.92\times$) and collapses to $0.40\times$ on CIC.

Conclusions. Under a controlled protocol the fidelity kernel does not outperform tuned classical baselines on either corpus. Its collapse on CIC is a property of the kernel rather than of the data, and kernel concentration is the mechanism.

1 Introduction

Malware family attribution is a multiclass problem over feature vectors that are high-dimensional, heavy-tailed and often near-degenerate. It has become a common testbed for quantum kernel methods, in which a quantum feature map replaces a classical kernel inside an otherwise standard support vector machine.

Claims of quantum competitiveness on such data are hard to evaluate, because three things routinely differ between the quantum and classical arms of a comparison: the rows each side is scored on, the dimensionality each side receives, and the metric used to report the outcome. Any of the three can manufacture or erase a difference.

This work asks one question under a protocol that removes those three degrees of freedom: *does a fidelity-kernel QSVM outperform tuned classical baselines at malware family attribution?* The contributions are:

1. a comparison protocol in which a single parameter fixes the classical feature count and the quantum qubit budget simultaneously, and in which the quantum arm’s subsample and cross-validation folds are persisted and replayed by the classical arm (§2.4);

2. results across two corpora, two tasks and three capacities, with paired significance tests on every comparable pair (§3);
3. evidence that one task widely used in this setting is saturated and therefore cannot rank methods at all (§3.1);
4. a diagnosis of the QSVM’s collapse on one corpus that separates a property of the data from a property of the kernel, using a classical control on the identical feature matrix (§3.6).

2 Materials and methods

2.1 Datasets

Two corpora, chosen so that no conclusion rests on a single feature representation. Each supports a binary detection task and a 15-class family attribution task.

CIC-MalMem-2022

Obfuscated-malware memory dumps described by 55 numeric forensic features covering process, handle, callback and DLL counts. The binary task uses all 58 596 rows and is balanced between benign and malware. The 15-class task drops benign rows, since benign detection is what the binary task measures, leaving 29 298 malware rows over 15 families with a $1.7\times$ ratio between the largest and smallest.

EMBER 2018

Static portable-executable features in 2381 dimensions. We work on an 18 014-row extract of the published test set; the full release is a single $200\,000 \times 2384$ block that cannot be read incrementally. The binary task uses all 18 014 rows and is 88.9% malware. The 15-class task keeps families with at least 500 samples, caps at the 15 largest, then downsamples each to the smallest kept count, giving 14 325 rows balanced at exactly 955 per family.

The two differ in kind as well as in size. CIC describes runtime behaviour and EMBER static file structure, and EMBER’s families are exactly balanced where CIC’s are mildly skewed. That contrast is used in §3.6.

2.2 Feature pipeline and the alignment parameter

Both frameworks consume the output of one pipeline: a zero-variance filter, a correlation filter at 0.95, standardisation, then PCA to $n_{\text{components}}$ components. The pipeline is fit on training-fold rows only.

$n_{\text{components}}$ is the single alignment point. It sets the number of features the classical models receive *and* the number of qubits the quantum circuit uses, so a comparison at $n_{\text{components}} = 6$ is six features against six qubits. Reporting a quantum model on six qubits against a classical model on all 2381 raw features would not be a comparison of methods.

2.3 Quantum model

The QSVM computes a fidelity kernel $K(x_i, x_j) = |\langle \phi(x_j) | \phi(x_i) \rangle|^2$ by encoding x_i , applying the inverse encoding of x_j , and reading the probability of the all-zeros measurement outcome. The resulting Gram matrix is passed to a support vector machine with a precomputed kernel. Two feature maps are compared: an angle encoding with a ring of entangling gates, and an IQP (ZZ-interaction) encoding. A bandwidth parameter scales inputs before rotation to counteract kernel concentration and defaults to $n_{\text{qubits}}^{-1/2}$.

All quantum results use a state-vector simulator.

2.4 Comparison protocol

Building the training Gram requires one circuit evaluation per *pair* of samples, so its cost is $O(n^2)$ in the number of training rows. Figure 1 measures the per-pair component of that cost.

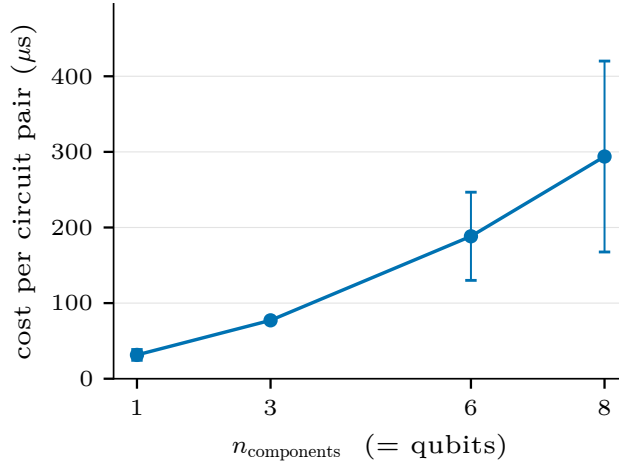


Figure 1: Cost of one circuit pair when building the training Gram, against qubit count, taken from a single session so the worker configuration is constant. Per-pair cost rises roughly tenfold from 1 to 8 qubits, and the number of pairs grows as n^2 , so the two effects compound.

Full-dataset quantum runs are therefore infeasible: at $n = 1000$ a single training fold already requires $\approx 320\,000$ circuit evaluations. Quantum results use a stratified 1000-row subsample. A separate sizing exercise fitted the scaling law and recommended $n = 900$ at a 60 s per-Gram budget, with a class-balance floor of $n \geq 750$ for the 15-class task; the 1000-row subsample sits just above that floor.

Because only the quantum arm is forced to subsample, the classical arm must be scored on the same rows or the comparison is void. The quantum runner writes its subsample indices and its five stratified outer folds to disk and the classical runner reloads them, so neither side chooses its own split. Each fold provides 800 training and 200 held-out rows.

2.5 Evaluation

All results are held-out macro-F1, the unweighted mean of per-class F1, averaged over the five outer folds. Hyperparameters are tuned by an Optuna inner loop inside each training fold, followed by one refit on the full training fold.

Accuracy is not used. On EMBER binary at $n_{\text{components}} = 1$ the QSVM attains 0.890 accuracy in three of five folds while the benign class’s F1 is exactly 0.000 in those same folds, because 88.9% of rows are malware and the model labels everything malware. Accuracy there reports near-success for a classifier that never identifies the class of interest.

Differences between a quantum and a classical model are assessed with a paired t -test over the five per-fold macro-F1 values.

2.6 Selection of comparable runs

Runs are grouped by cross-validation sweep rather than by configuration. The same configuration was swept more than once over the course of the work, so a parameter-based grouping returns 10, 15 or 20 records for a five-fold sweep and averages distinct hyperparameter searches together; 57 parameter groups in the experiment log have a size other than five. For the QSVM there is a second reason: the encoding is a *per-fold outcome* rather than a configuration key, because the tuner selects it inside each fold, so a single sweep tuning over both encodings can appear as four angle folds and one IQP fold.

Fold records are therefore grouped by parent run, retaining only sweeps that completed with exactly five folds. Under that rule the log holds 125 nested sweeps, 123 of them complete.

3 Results

3.1 The binary task on CIC-MalMem is saturated

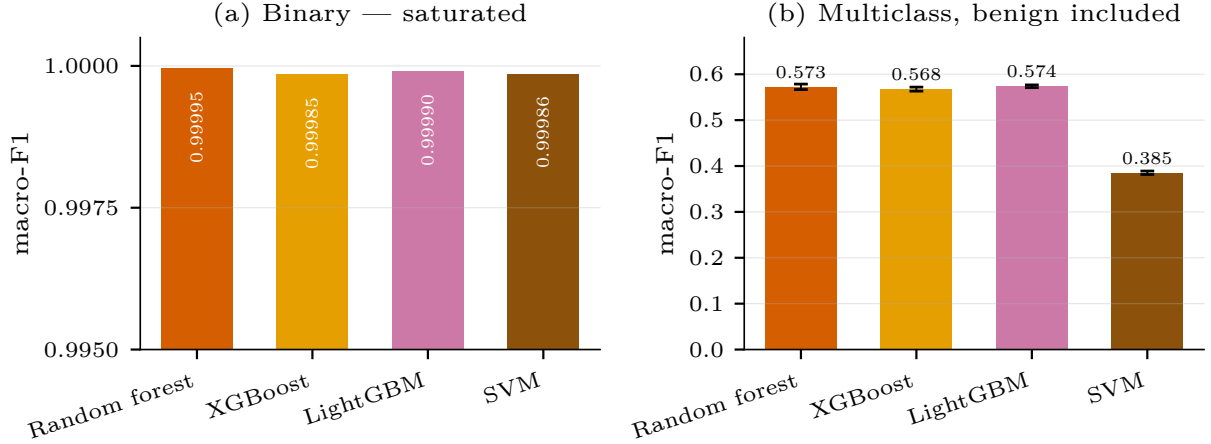


Figure 2: Four classical models on the full CIC-MalMem dataset without dimensionality reduction. (a) On the binary task all four exceed 0.9998 macro-F1 and the spread between best and worst is 10^{-4} . (b) On a multiclass formulation that still included benign as a class, the same models separate by 0.19 macro-F1.

Figure 2(a) shows that the binary task on CIC-MalMem leaves no headroom in which to measure a difference. Any ranking of methods on it measures noise.

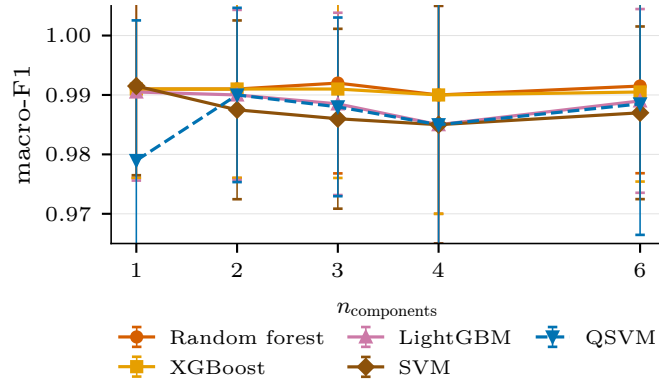


Figure 3: CIC-MalMem binary on the 1000-row subsample across the PCA sweep. Every model, quantum and classical, lies between 0.977 and 0.992 macro-F1, and every difference is inside the fold-to-fold error bars. Aggregated over all recorded folds, since individual sweeps are not separable for this configuration (§4.2).

The saturation survives dimensionality reduction and subsampling (Figure 3). The QSVM does draw level with the classical models here, at a gap of 0.006 macro-F1. That is a property of the task rather than evidence of parity between methods, and CIC binary is treated as a control throughout.

Table 1: EMBER 2018, held-out macro-F1, mean over five outer folds. Best classical model in bold.

	Random forest	XGBoost	LightGBM	SVM	QSVM angle	QSVM IQP
<i>Binary</i>						
$n_{\text{components}} = 1$	0.5582	0.5451	0.5093	0.5314	0.4532	0.4587
$n_{\text{components}} = 3$	0.6575	0.6465	0.6434	0.6306	0.4924	0.5084
$n_{\text{components}} = 6$	0.6835	0.6680	0.6593	0.6724	0.5138	0.5137
<i>15-class</i>						
$n_{\text{components}} = 1$	0.5562	0.5313	0.5410	0.5294	0.1688	0.1541
$n_{\text{components}} = 3$	0.7194	0.7089	0.7035	0.7389	0.3908	0.4797
$n_{\text{components}} = 6$	0.7903	0.7659	0.7709	0.7930	0.4232	0.5195

3.2 Classical models lead on every discriminating configuration

Table 1 gives the EMBER results. Which classical model leads depends on the task: random forest on binary at every capacity, SVM on 15-class at $n_{\text{components}} \in \{3, 6\}$. Comparing the QSVM against SVM alone would understate the classical advantage on binary by 0.011 to 0.027 macro-F1, which is why all four baselines are reported.

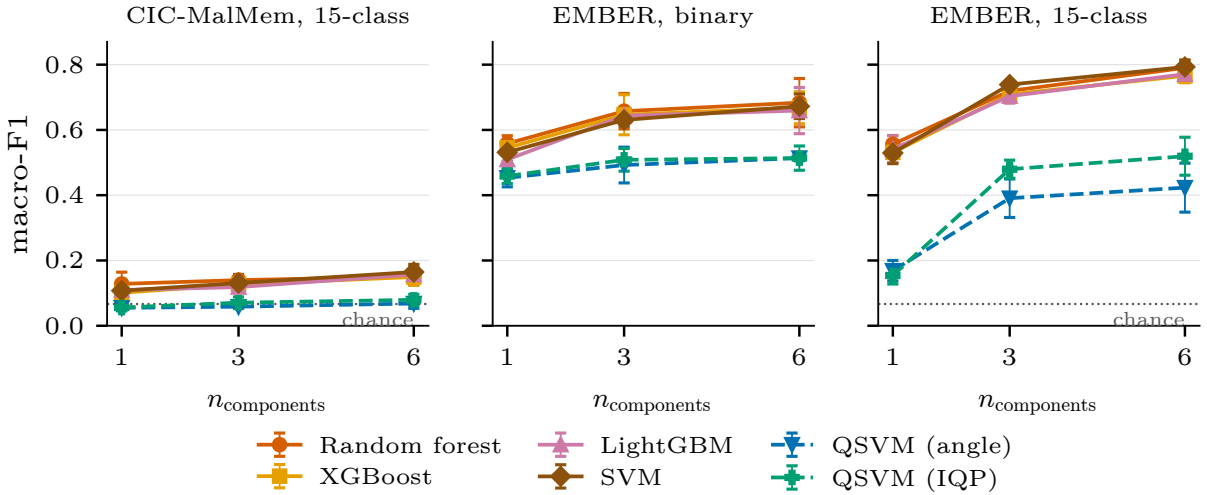


Figure 4: Macro-F1 against the shared feature and qubit budget. Solid lines are classical, dashed lines the two QSVM encodings. Error bars are one standard deviation over the five outer folds, and the dotted line marks the 15-class chance level.

Figure 4 shows behaviour across capacity. Both frameworks improve monotonically but at different rates: on EMBER binary the QSVM gains 0.055 macro-F1 from one to six components where the random forest gains 0.125, so the gap widens with capacity rather than closing. On CIC 15-class the quantum curves sit on the chance level at every capacity tested.

Summarising (Figure 5): the classical lead is 0.085 on CIC 15-class, 0.170 on EMBER binary and 0.274 on EMBER 15-class. On the saturated control the two are 0.006 apart.

3.3 The differences are statistically significant

All 84 comparable pairs favour the classical model and all 84 reach $p < 0.05$ on the paired t -test. The largest p -value across every pair is 0.0241 and the smallest is 4×10^{-6} .

A Wilcoxon signed-rank test returns exactly 0.0625 for all 84 pairs. With five paired samples that is the smallest attainable two-sided p -value, so the test cannot reach 0.05 at this fold count regardless of effect size; all 84 reaching the floor indicates that every pair achieved the most extreme rank configuration available.

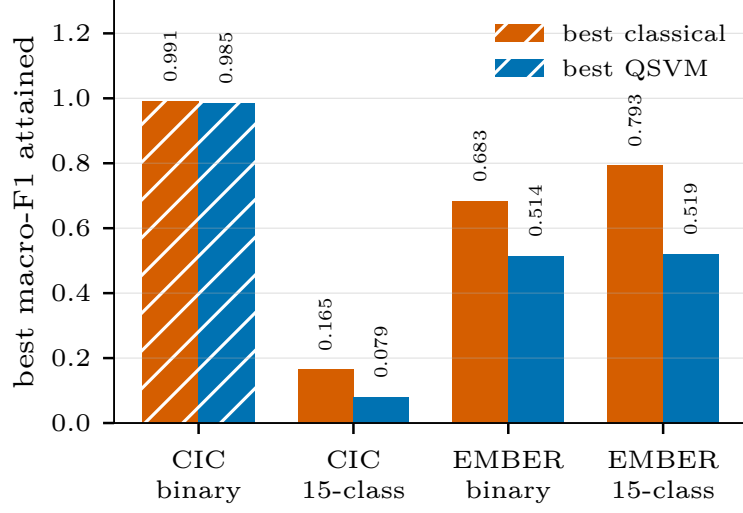


Figure 5: Best classical against best QSVM in each corpus and task, over $n_{\text{components}} \in \{1, 3, 6\}$. Hatched bars are aggregated from raw folds because individual sweeps are not separable for that configuration.

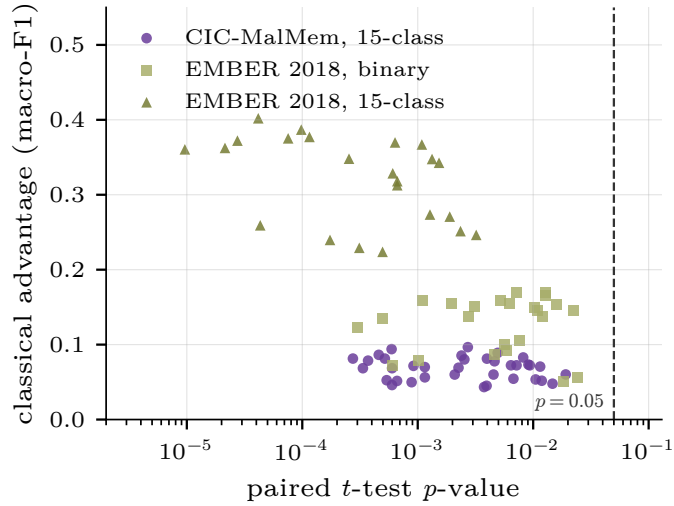


Figure 6: Effect size against paired t -test p -value for all 84 comparable quantum-classical pairs. Every point lies left of $p = 0.05$ and above zero.

3.4 The encoding advantage is corpus-dependent

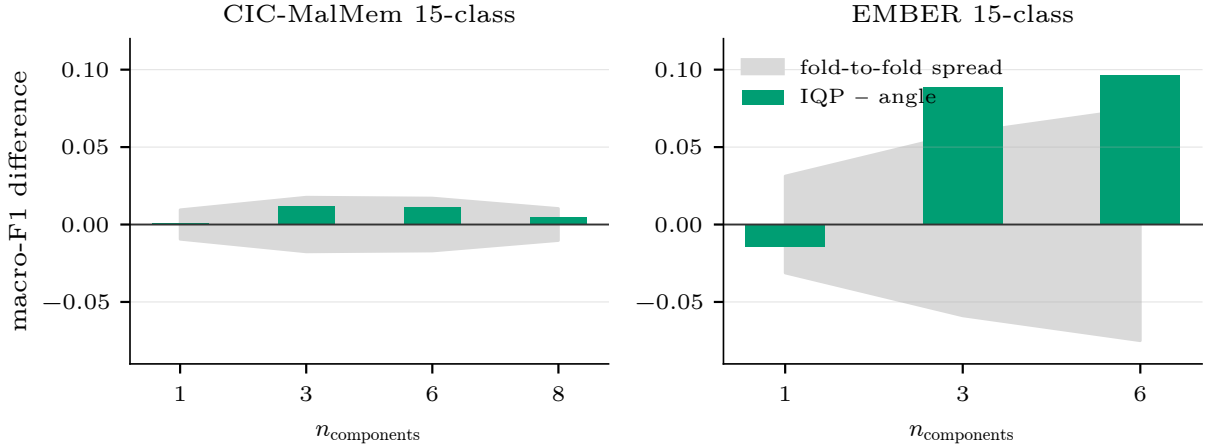


Figure 7: IQP minus angle encoding on the 15-class task. The grey band is the fold-to-fold spread of the encodings themselves, so a difference inside that band is within noise.

IQP is never worse than angle, but the margin is resolvable on only one corpus (Figure 7). On CIC the differences are +0.0009, +0.0122, +0.0113 and +0.0046 at $n_{\text{components}} = 1, 3, 6, 8$, each smaller than either encoding’s own standard deviation. On EMBER the differences at $n_{\text{components}} = 3$ and 6 are +0.0889 and +0.0963, well outside the corresponding spread. Extending CIC 15-class to 8 qubits did not lift performance above chance, which excludes qubit count as the limiting factor there.

3.5 The subsample preserves its population

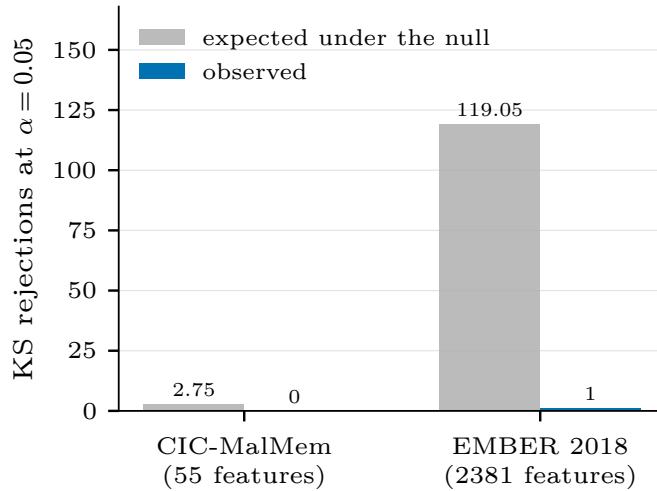


Figure 8: Per-feature two-sample Kolmogorov–Smirnov rejections at $\alpha = 0.05$, 1000-row subsample against full population, beside the count chance alone produces.

Since every quantum result is computed on 1000 rows, the subsample’s fidelity to its population is a precondition for the rest. A rejection count is interpretable only against its null expectation, because a faithful subsample still rejects about $\alpha \cdot n_{\text{features}}$ features by chance. CIC rejects 0 of 55 features where ≈ 2.75 were expected, and EMBER 1 of 2381 where ≈ 119 were expected (Figure 8). Maximum class-proportion drift is below 0.001 on both.

3.6 Kernel diagnostics

The QSVM reaches 0.056–0.079 on CIC 15-class against a chance level of ≈ 0.067 , while the same method reaches 0.42–0.52 on EMBER 15-class. Class imbalance is excluded, since EMBER’s families are exactly balanced and CIC’s only $1.7\times$ skewed, and qubit count is excluded by the preceding section. Two further measurements separate a property of the data from a property of the kernel.

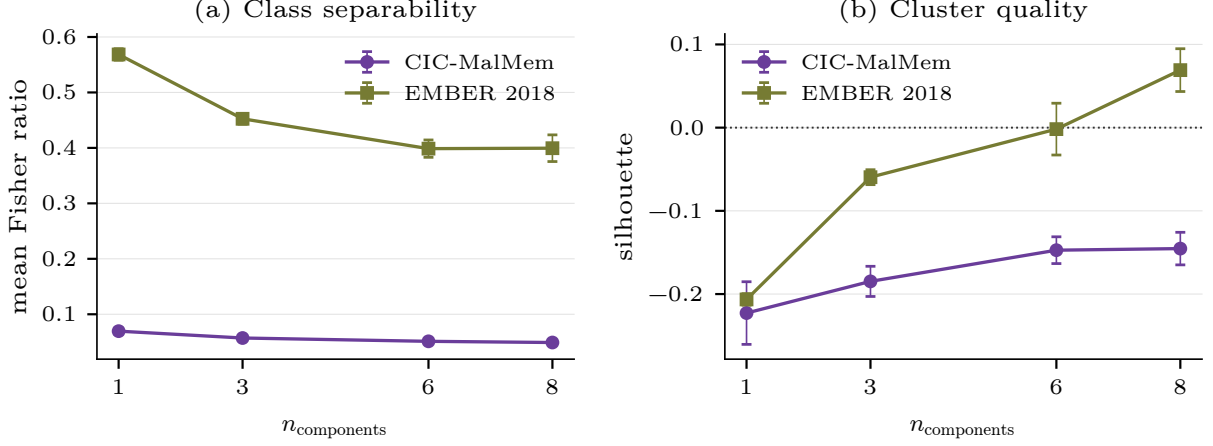


Figure 9: Class separability of the shared PCA projection, averaged over the five outer folds. Both metrics are scale-invariant and therefore comparable across corpora of very different dimensionality.

Feature geometry. On the identical projected features, EMBER’s families are $\approx 8\times$ more separable by one-vs-rest Fisher ratio, and up to $50\times$ on the worst-case family. CIC’s silhouette coefficient is negative at every $n_{\text{components}}$: the average point lies closer to another family’s centroid than to its own. CIC’s PCA nonetheless explains substantially more variance (33–81% against 8–30%), so its variance is concentrated in directions that do not discriminate family.

This accounts for the poor absolute performance of every method on CIC 15-class. It does not account for the QSVM-specific deficit, since a classical SVM extracts 0.108–0.165 macro-F1 from these same features where the QSVM extracts 0.056–0.079.

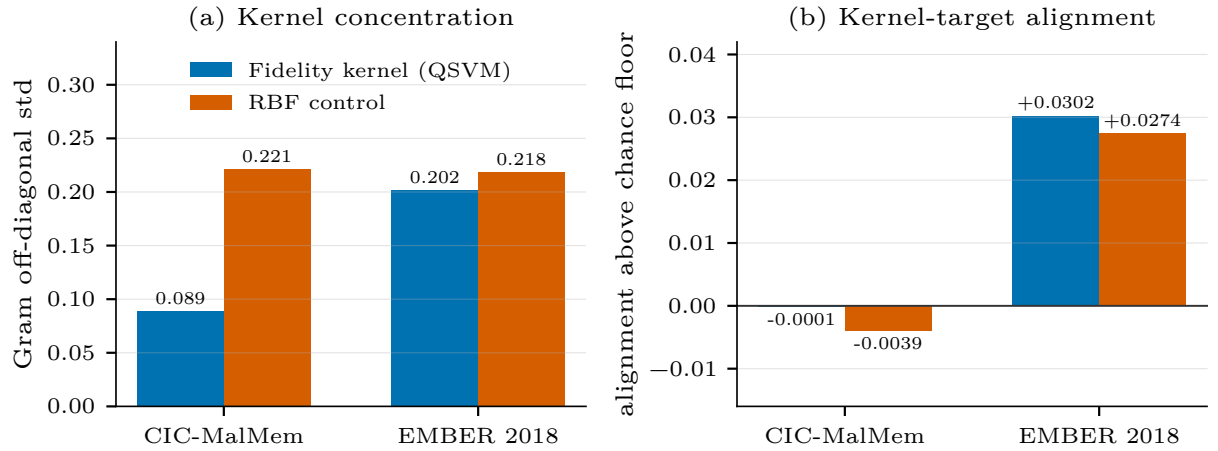


Figure 10: First outer training fold, $n_{\text{components}} = 6$, IQP encoding, 800 rows, with an RBF control built on the identical feature matrix. (a) Off-diagonal Gram spread. (b) Kernel-target alignment relative to its floor: a constant kernel scores $1/\sqrt{15} \approx 0.258$ on 15 balanced classes, so only the excess is informative.

Kernel geometry. The RBF control produces near-identical off-diagonal spread on both corpora (0.2212 against 0.2184), so CIC’s features do not cause a classical kernel to concentrate. The fidelity kernel tracks RBF on EMBER at $0.92\times$ and collapses to $0.40\times$ on CIC: every off-diagonal similarity is compressed into a band $2.5\times$ narrower from the same inputs, driving the Gram toward a scaled identity.

On alignment, the CIC fidelity kernel lands 0.000069 *below* the constant-kernel floor, making it indistinguishable from a matrix of ones, and the RBF control is also below it. On EMBER both clear the floor by ≈ 0.03 . Alignment therefore separates the two corpora, and concentration separates the two kernels.

Across the full sweep the CIC fidelity Gram is $1.9\text{--}2.6\times$ less spread than EMBER’s at every capacity and for both encodings, and concentration worsens with qubit count on both, which is the trend the bandwidth parameter exists to counteract.

4 Discussion

4.1 Interpretation

The central result is negative and consistent. Under a protocol that equalises rows, folds, dimensionality and metric, the fidelity-kernel QSVM does not outperform tuned classical baselines on either corpus or either task, and the deficit is significant on every comparable pair.

The one configuration in which the two are level is the saturated CIC binary task, where all methods sit within 0.015 macro-F1 of one another. Reports of quantum parity on binary malware detection should be read with that ceiling in mind: a task on which four tuned classical models differ by 10^{-4} cannot distinguish a quantum kernel from a classical one either.

The more transferable finding concerns *why* the QSVM fails on one corpus and not the other, and the RBF control isolates it. Since a classical kernel on the identical feature matrix does not concentrate, the collapse is not a consequence of CIC’s features being hard; it is a consequence of the fidelity kernel’s behaviour on those features. A Gram matrix approaching a scaled identity presents the support vector machine with a geometry in which all pairs are equally similar, which is what a macro-F1 pinned to chance looks like internally. This predicts that the method will struggle wherever the encoded states become mutually near-orthogonal, independently of the corpus.

4.2 Limitations

Scale

Every quantum result derives from 1000 rows, forced by the quadratic kernel cost, against source corpora of 29 298 and 14 325 rows for the 15-class tasks. The subsample is verified faithful, but the reported standard deviations are over five folds of 200 held-out rows.

Bandwidth is untested on CIC

Concentration is the identified mechanism and bandwidth is the parameter that acts on it, but no bandwidth sweep was run on CIC. §3.6 identifies a mechanism; it does not demonstrate a remedy.

Alignment is a weak predictor

Alignment reports that neither kernel carries label information on CIC, yet a classical SVM reaches 0.16 there. Alignment is a global, unweighted average over all pairs, and a support vector machine requires no such global property, only support vectors, a regularisation constant and class weights. It ranks corpus difficulty usefully and predicts attainable SVM performance poorly. The concentration measurement does not share this weakness, since it describes the matrix the SVM actually receives.

One quantum method

The comparison is a fidelity-kernel QSVM against classical baselines, not quantum methods in general.

Simulator only

All quantum results use a state-vector simulator, with no hardware execution and no noise model. Concentration behaviour on hardware, where sampling noise adds its own floor, is untested.

One configuration is not separable

Every CIC binary run, classical and quantum alike, predates sweep-level tracking in the experiment log and carries no recoverable sweep boundary; 45 fold records per model resolve into two distinct sweeps per $n_{\text{components}}$ with no identifier separating them. Those records are valid measurements and are used descriptively in Figures 3 and 5, but no grouping was reconstructed, since the time gaps between runs are not a recorded boundary. The limitation is symmetric across frameworks.

No sample-level test between frameworks

McNemar’s test requires per-sample predictions from both models on a shared test set. QSVM predictions were not persisted during the reported sweeps, so cross-framework evidence rests on five paired fold-level scores rather than on paired sample-level outcomes.

Fold count

Five outer folds support the paired t -test and are too few for Wilcoxon, whose minimum attainable p -value at $n = 5$ is 0.0625. Six folds would place 0.03125 within reach.

4.3 Future work

Two steps follow directly from §3.6. First, sweep bandwidth on CIC: the mechanism is identified and the corresponding parameter has never been tuned there beyond its default. Second, move to six outer folds, which makes a non-parametric test usable and removes the ceiling noted above.

5 Conclusion

Under a protocol that holds rows, folds, dimensionality and metric fixed across both arms, a fidelity-kernel QSVM does not outperform tuned classical baselines at malware family attribution on either of two corpora. Classical models lead on every configuration where the task can discriminate at all, by 0.085 to 0.274 macro-F1, with $p < 0.05$ on all 84 comparable pairs. The single level result occurs on a task where all methods are saturated and no method can be ranked.

The QSVM’s collapse to chance on one corpus is attributable to kernel concentration rather than to feature difficulty: a classical RBF kernel on the identical feature matrix retains its off-diagonal spread while the fidelity kernel falls to $0.40\times$ of it. That is a statement about the method rather than about the data, and it identifies both the failure mode and the parameter that acts on it.